Note that the exponent of charge q (source) and exponent of electric field E is unity. Similarly, the exponent of current element Idl (source) and exponent of magnetic field B is unity. In other words, electric field $\vec{E}$ is proportional only to charge (source) and not on higher powers of charge (q^2, q^3, etc). Similarly, magnetic field $\vec{B}$ is proportional to current element $Id\vec{l}$ (source) and not on square or cube or higher powers of current element. The cause and effect have linear relationship.

3.8.2 Magnetic field due to long straight conductor carrying current

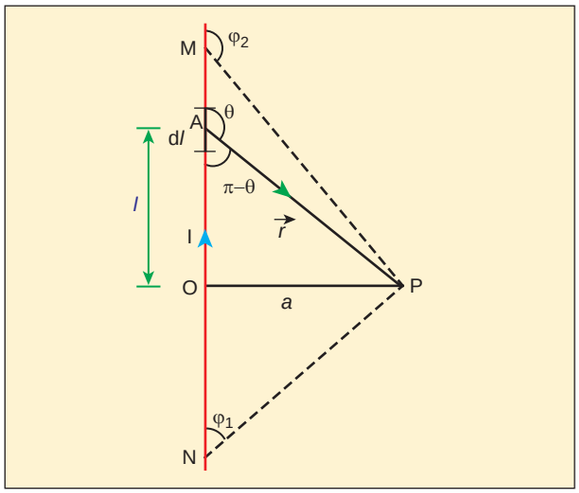


Figure 3.39 Magnetic field due to a long straight current carrying conductor

Consider a long straight wire NM with current I flowing from N to M as shown in Figure 3.39. Let P be the point at a distance a from point O. Consider an element of length dl of the wire at a distance l from point O and $\vec{r}$ be the vector joining the element dl with the point P. Let θ be the angle between $d\vec{l}$ and $\vec{r}$. Then, the magnetic field at P due to the element is

$$d\vec{B} = \frac{\mu_\circ I}{4\pi}\frac{d\vec{l}}{r^2}\sin\theta \begin{pmatrix} unit\ vector\ perpendicular \\ to\ d\vec{l}\ and\ \vec{r} \end{pmatrix}$$

The direction of the field is perpendicular to the plane of the paper and going into it. This can be determined by taking the cross product between two vectors $d\vec{l}$ and $\vec{r}$ (let it be $\hat{n}$). The net magnetic field can be determined by integrating equation (3.38) with proper limits.

From the Figure 3.39, in a right angle triangle PAO,

$$\tan(\pi-\theta) = \frac{a}{l}$$

$$l = -\frac{a}{\tan\theta} \quad (\text{since } \tan(\pi-\theta) = -\tan\theta)$$

$$l = -a\cot\theta \quad and \quad r = a\,cosec\,\theta$$

Differentiating,

$$dl = a\,cosec^{\,2}\theta\, d\theta$$

$$d\vec{B} = \frac{\mu_\circ I}{4\pi}\frac{\left(a\,\text{cosec}^{\,2}\theta\, d\theta\right)}{\left(a\,\text{cosec}\,\theta\right)^2}\sin\theta\, d\theta\,\hat{n}$$

$$d\vec{B} = \frac{\mu_\circ I}{4\pi}\frac{\left(a\,\text{cosec}^{\,2}\theta\, d\theta\right)}{a^2\,\text{cosec}^2\,\theta}\sin\theta\, d\theta\,\hat{n}$$

$$= \frac{\mu_\circ I}{4\pi a}\sin\theta\, d\theta\,\hat{n}$$

This is the magnetic field at a point P due to the current in small elemental length. Note that we have expressed the magnetic field OP in terms of angular coordinate i.e. θ. Therefore, the net magnetic field at the point P which can be obtained by integrating $d\vec{B}$ by varying the angle from $\theta = \varphi_1$ to $\theta = \varphi_2$ is

$$\vec{B} = \frac{\mu_\circ I}{4\pi a}\int_{\varphi_1}^{\varphi_2}\sin\theta\, d\theta\,\hat{n} = \frac{\mu_\circ I}{4\pi a}(\cos\varphi_1 - \cos\varphi_2)\,\hat{n}$$

For a an infinitely long straight wire, $\varphi_1 = 0$ and $\varphi_2 = \pi$, the magnetic field is

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